

Chapter 2: 2-Tier and 3-Tier Architecture

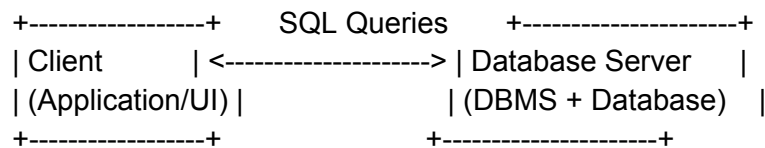
What is DBMS Architecture?

DBMS architecture defines **how users interact with the database** and **how data flows between the user, application, and database**.

It organizes the system into different **layers (tiers)** to improve security, scalability, and maintainability.

1. Two-Tier (2-Tier) Architecture

In a **2-tier architecture**, the client (user application) communicates **directly** with the database server.



Components

1. Client

- User Interface (UI)
- Sends SQL queries
- Displays results

Examples:

- Banking software
- College desktop application
- Library management software

2. Database Server

- Stores the database
 - Executes SQL queries
 - Returns the requested data
-

Working

Step 1

User enters information.

Example:

Login

Username: Rahul

Password: ****

↓

Step 2

The application sends an SQL query to the database.

```
SELECT * FROM Student  
WHERE RollNo = 101;
```

↓

Step 3

The DBMS processes the query.

↓

Step 4

The result is returned to the application.

Name : Rahul

CGPA : 8.5

Department : E&TC

Advantages

- Simple architecture
- Easy to develop
- Faster communication (direct connection)
- Lower cost
- Suitable for small organizations

Disadvantages

- Poor security (client connects directly to the database)
- Difficult to maintain for large applications
- Limited scalability
- Heavy load on the database server

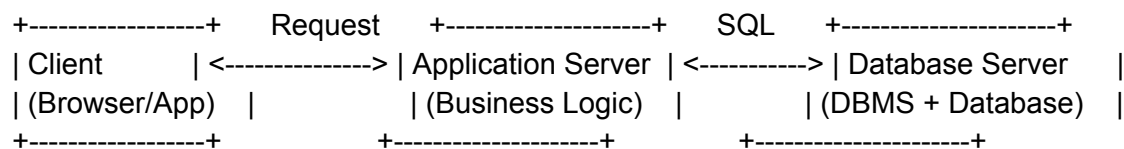
Real-Life Examples

- Desktop applications
- Microsoft Access
- Local payroll software
- Small office inventory system

2. Three-Tier (3-Tier) Architecture

In a **3-tier architecture**, an **Application Server (Business Logic Layer)** is placed between the client and the database.

Clients **cannot access the database directly**.



Components

1. Presentation Layer (Client)

Responsible for:

- User Interface
- User interaction
- Displaying information

Examples:

- Mobile App
 - Website
 - Desktop GUI
-

2. Application Layer (Business Logic)

This is the **brain of the application**.

Responsibilities:

- Authentication
- Validation
- Business rules
- Processing requests
- Communicating with the database

Example:

User wants to transfer ₹5000.

The application checks:

- Is the user logged in?
- Is the balance sufficient?
- Is the amount valid?

Only then is the SQL query sent to the database.

3. Database Layer

Responsible for:

- Storing data
- Retrieving data

- Executing SQL queries
 - Backup and recovery
-

Working of 3-Tier Architecture

Step 1

User sends a request.

Login

↓

Step 2

Application server verifies:

- Username
- Password
- Permissions

↓

Step 3

Application sends SQL query.

```
SELECT * FROM Users  
WHERE Username='Rahul';
```

↓

Step 4

Database returns data.

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Step 5

Application formats the response.

↓

Step 6

Client displays the result.

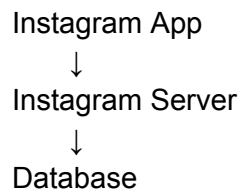
Real-Life Example

Amazon

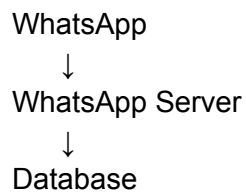


The customer never communicates directly with the database.

Instagram



WhatsApp



Why Do We Need the Application Layer?

Without it:

User

↓
Database

Anyone who knows SQL could directly manipulate the database.

Example:

```
DELETE FROM Student;
```

This could delete all records if proper controls are not in place.

With an application layer:

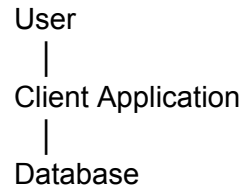
- Only authorized operations are allowed.
 - User input is validated.
 - Security rules are enforced.
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Difference Between 2-Tier and 3-Tier Architecture

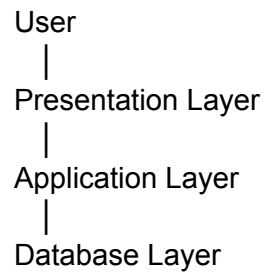
Feature	2-Tier	3-Tier
Number of Layers	2	3
Client Connection	Direct to Database	Through Application Server
Business Logic	Client	Application Server
Security	Lower	Higher
Scalability	Limited	High
Performance	Good for small systems	Better for large systems
Maintenance	Difficult	Easier
Cost	Lower	Higher
Suitable For	Small applications	Enterprise applications

Comparison Diagram

2-Tier



3-Tier



Examples

2-Tier

- Microsoft Access
 - Small College Management System
 - Local Billing Software
 - Desktop Inventory System
-

3-Tier

- Amazon
 - Flipkart
 - Instagram
 - Facebook
 - WhatsApp
 - Netflix
 - Banking Systems
 - Hospital Management Systems
-

Placement Interview Questions

Q1. What is DBMS architecture?

DBMS architecture defines how users, applications, and the database interact to process and manage data efficiently.

Q2. What is a 2-tier architecture?

A 2-tier architecture consists of a client directly communicating with the database server without an intermediate application server.

Q3. What is a 3-tier architecture?

A 3-tier architecture separates the system into three layers: Presentation Layer (client), Application Layer (business logic), and Database Layer (DBMS). The client communicates with the database through the application server.

Q4. Which architecture is more secure?

3-tier architecture, because the client cannot directly access the database.

Q5. Which architecture is used by modern web applications?

3-tier architecture is widely used in web and enterprise applications due to its better security, scalability, and maintainability.